Ali Hamza Malik

ahmalik@umass.edu | alihamzamalik.me | linkedin.com/alihamzamalik | github.com/alirafiqmalik

Skills

Programming Languages: Python, C/C++, Julia, Rust, JavaScript, Kotlin, MATLAB
Formal Methods: Symbolic Execution, Linear Temporal Logic (LTL), Finite State Automata (FSA), Model Checking, SMT encoding, Model Checking (TLA+, nuXMV, STORM), Theorem Proving (Coq, Lean)
Generative AI/ML: Huggingface, Transformers, PyTorch, TensorFlow, Keras, SFT, DPO, RLHF, RLAF (RL-Automated Feedback), LoRA, QLoRA, RAG, Specification-Aware Fine-Tuning, Agentic AI
DevOps & Cloud: Git/GitHub, Docker, Shell Scripting, CMake, AWS (EC2, Route 53), MongoDB, N8N

Relevant Experience

Graduate Research Assistant, Khwarizmi Lab September 2023 – Present
University of Massachusetts Amherst

- Discovered previously undocumented security flaws in quantum key distribution (QKD) protocols under quantum-classical interactions by developing a Tamarin prover-based verification framework
- Uncovered 3 critical access control vulnerabilities in U.S. ACH direct payment systems by formally modeling banking payment protocols in TLA+ to perform bounded model checking

Undergraduate Research Assistant, Communication Systems and Networks Lab September 2022 – July 2023
National University of Sciences & Technology

- Built an event-driven coordination system using C++ and MAVLink for multi-UAV aerial swarms on Raspberry Pi and Pixhawk hardware stack to enable autonomous multi-agent mission planning
- Optimized leader–follower formation control algorithms (flock, line, helical) with dynamic reconfiguration, reducing formation-switching latency to under 2 minutes

Hardware Security Intern, IC Design Lab June 2022 – September 2022
National University of Sciences & Technology

- Developed ENIGMA, an automated Python framework for inserting logic-locking defenses into Verilog HDL designs for reducing IP reverse-engineering risk
- Developed SAT-based attack algorithms that successfully recovered logic-locking keys from obfuscated circuits in under 6 hours to expose weaknesses in 2 commercial logic-locking schemes

Machine Learning Intern, TUKL Deep Learning Lab June 2021 - September 2021
National University of Sciences & Technology

- Built an automated ETL pipeline using Python and regex to process 10,000+ PDF raw court documents to fine-tune NLP transformer models (BERT and RoBERTa) for verdict prediction of court cases

Education

PhD in Electrical & Computer Engineering Expected May 2028
University of Massachusetts Amherst

Teaching Assistant: ECE 304: Junior Design Project, ECE 361: Fundamentals of Electrical Engineering

BE in Electrical Engineering *and* Minor in Computer Science Completed June 2023
National University of Sciences and Technology

Capstone Project: Optimized security-cost tradeoffs of hardware obfuscation on a proprietary RISC-V microprocessor
Funded Project: Grant-funded research implementing a low-latency control system for multi-UAV aerial swarms

Relevant Projects

LogicTune — *Python, PyTorch, LLM, RLAF, DPO, LoRA, Model Checking*

- Improved autonomous driving safety from 60% to 90% with DPO and feedback from model checking.

Voice-Controlled Home Automation System — *C/C++, Python, JavaScript/HTML, PyTorch, Tensorflow*

- Developed an ESP32-based voice automation system with on-device speech recognition

Honors and Awards

2023 - Rector's Gold Medal: Awarded for best senior project, National University of Sciences & Technology
2022 - 2nd Place, CSAW'22 LLC: For a global security hackathon competition, NYU School of Engineering